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Interaction of *Pseudomonas aeruginosa* and *Staphylococcus aureus* with *Listeria innocua* in dual species biofilms and inactivation following disinfectant treatments

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ABSTRACT

Biofilms pose a serious hygienic concern in the food processing environment due to the strong antimicrobial tolerance and thus higher risk of food contamination. The aim of this study was to investigate dual species biofilm formation of *P. aeruginosa* ATCC 27853 or *S. aureus* FI 2 with *L. innocua* FI 2 and biofilm inactivation using QAC-, tertiary alkyl amine-, and chlorine-based disinfectants; the effect of biofilm maturity and comparison with single species biofilms are reported. We used epifluorescence microscopy (EFM) coupled with fluorescence *in situ* hybridization (FISH) and LIVE/DEAD staining to evaluate species contribution and cell inactivation; cells were prepared using the LabTek Chamber Slide System. In dual species biofilms during 72-h incubation time, *P. aeruginosa* produced a higher and *S. aureus* slightly lower cell numbers than the companion species of *L. innocua*. In dual species biofilm inactivation experiments, *P. aeruginosa* was more resistant to disinfectant treatment of 24-h compared to 72-h; however, *S. aureus* was more resistant at 72-h compared to 24-h. In addition, *P. aeruginosa* was more resistant to tertiary alkyl amine- and chlorine-based disinfectants and *S. aureus* was more resistant to all three disinfectants in dual species biofilms compared to single species biofilms. In the LabTek™ experiment, when nutrients were repeatedly provided, FISH showed almost equal contribution of *P. aeruginosa* and *L. innocua* into the biofilm as well as a predominance of *S. aureus* when grown with *L. innocua*. Regardless of biofilm composition, the LIVE/DEAD staining revealed that the majority of cells had damaged membranes after the treatment with the QAC-based disinfectant, while approximately half of the cells remained intact following the treatment with the tertiary alkyl amine-based disinfectant. As for chlorine, cells in *P. aeruginosa-L. innocua* biofilm were mostly dead whereas cells in *S. aureus-L. innocua* biofilm were largely viable, and thus were protected under the applied conditions. This study shows that dual species biofilms with foodborne pathogens are more resistant to disinfectant treatment than their single species counterparts and that biofilm maturity as well as environmental conditions, i.e. access to nutrients, play an important role in their response to the treatments tested.

1. Introduction

Bacterial biofilms, especially those produced by foodborne pathogens, are challenging to the food industry as they pose a risk for food contamination (Le Thi; Prigent-Combaret, Dorel, & Lejeune, 2001; Simões, Simões, & Vieira, 2010). Cleaning and disinfection procedures are designed to ensure appropriate hygienic conditions in food processing plants. Microorganisms capable of both adhering to abiotic surfaces and producing biofilms are a recurrent concern, because their biofilms form a diffusion barrier preventing antimicrobial penetration into the biofilm (Berrang, Meinersmann, Frank, Smith, & Genzlinger, 2005; Chaitiemwong, Hazeleger, & Beumer, 2010). It has been demonstrated in multiple studies that biofilms are more resistant to many processes such as drying and UV radiation and also to antimicrobial agents such as antibiotics, surfactants or disinfectants compared to their planktonic counterparts (Aarnisalo, Lunden, Korkeala, & Wirtanen, 2007; Gram, Bagge-Ravn, Ng, Gymoese, & Vogel, 2007; Minei, Gomes, Regianne, D' Angelis, & De Martinis, 2008; Kim; Pitts, Stewart, Camper, & Yoon, 2008; Amalaradjou, Norris, & Venkitanarayanan, 2009; Cruz & Fletcher, 2011). Members of the genus *Pseudomonas* are well known participants of biofilms formed on food contact surfaces and often serve as model organisms for biofilm studies (Giaouris, Chorianopoulos, Doulgeraki, & Nychas, 2013; Sharma et al., 2014; Wang, Cai, Li, Xi, & Zhou, 2018). They have been shown to

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colonize various surfaces (Giaouris et al., 2013; Olszewska, Kocot, Stanowicka, & Łaniewska Trokenheim, 2016; Saá Ibusquiza, Cabo, & Herrera, 2010), develop a 3-D 'mushroom-like' structure (Sharma et al., 2014), and interact with other species to form mixed species biofilms (Saá Ibusquiza et al., 2010). Other microbes able to form biofilms on different surfaces found in food processing environments belong to the *Staphylococcus* genus, with *Staphylococcus aureus* being the most common causative agent of food poisoning (Galié, García-Gutiérrez, Miguélez, Villar, & Lombó, 2018). Moreover, resistance of *S. aureus* to disinfectants has been reported to increase in biofilms (Campana, Ciandrini, & Baffone, 2018; Neopane, Nepal, Shrestha, Uehara, & Abiko, 2018). Intense research is also underway on *Listeria monocytogenes*, particularly on its capability for biofilm formation on various surfaces and under different conditions or on the effectiveness of its inactivation in biofilms (e.g., Doijad et al., 2015; Olszewska, Zhao, & Doyle, 2016). The majority of investigations addressing biofilms concern single species biofilms, while in both the natural and food processing environments biofilms are often composed of multiple species (Reij & Den Aantrekker, 2004). In the dual and mixed species biofilms, microbial interactions can have either a synergistic or a deleterious effect resulting in their lower or higher resistance to various physical and chemical factors (Boari, Alves, Reis, Savian, & Piccoli, 2009; Christensen, Haagensen, & Molin, 2002; Rao, Webb, & Kjelleberg, 2005; Rodríguez-López, Carballo-Justo, Draper, & Cabo, 2017; Van der Veen & Abee, 2011). Despite the above mentioned studies, little is known about how interactions between microorganisms in mixed species biofilms contribute to the resistance of biofilms to environmental, physical, and chemical stresses. It is well accepted that a biofilm is an architecturally-complex community, which changes dynamically throughout its life cycle depending on environmental factors (Kocot & Olszewska, 2017), which may affect the efficacy of disinfectant treatments used in the food industry.

Acquiring knowledge on biofilms and their thorough analysis are feasible owing to the progress in microscopic techniques. Particularly useful techniques in biofilm analysis include: scanning electron microscopy (SEM), epifluorescence microscopy (EFM), and confocal microscopy (CM), among which EFM is widely used as it enables the multi-parametric analysis of cells depending on the adopted protocol (Joux & Lebaron, 2000). Its applicability is largely dependent on the fluorescent markers used, which may inform about membrane integrity, enzymatic activity or membrane potential of the cells (Kocot & Olszewska, 2017; Plichova et al., 2014). For instance, Olszewska, Kocot, et al. (2016) used EFM to study adherence stages of single species biofilm of *P. aeruginosa,* whereas Rodríguez-López et al. (2017) employed EFM to analyze changes in the viability of *L. monocytogenes-Escherichia coli* in mixed species biofilm after sublethal exposures to pronase (PRN) and benzalkonium chloride (BAC). An advantage of EFM is also the possibility of biofilm analysis after genetic modification of cells, e.g. GFP- or RFP-labelled strains (Azeredo et al., 2016). Rodríguez-López et al. (2017) examined a mixed species biofilm of *L. monocytogenes-E. coli* cells modified for GFP and mCherry constitutive expression in the context of biofilm removal using combined enzyme-BAC treatments. Another advantage offered by EFM is the possibility of using oligonucleotide probes fluorescentically labelled with e.g. Cy3 (cyanine 3), Cy5 (cyanine 5) or fluorescein, which may be especially useful in analyses of the mixed species biofilms (Stoecker, Dorninger, Daims, & Wagner, 2010; Valm, Welch, Rieken, & Hasegawa, 2011). For instance, fluorescence *in situ* hybridization (FISH) was applied by Garcia-Almendarez, Cann, Martin, and Guerrero-Legarreta (2008) to study *Lactococcus lactis-L. monocytogenes* mixed species biofilm using probes labelled with fluorescein and Cy3, respectively, and enabled observing a higher count of *L. lactis*.

Therefore, the aim of this study was to elucidate the species performance and effects of different disinfectants on biofilm-producing pathogenic bacteria, *Pseudomonas aeruginosa* and *Staphylococcus aureus* in dual species biofilms with *Listeria innocua* as a companion species depending on their stage of maturity and in comparison to their single species biofilms. *L. innocua* was selected based on a study where it demonstrated good capability to produce biofilms (data not shown). *L. innocua* is genetically, ecologically, and phenotypically close to *L. monocytogenes* and thus a good surrogate which frequently enters the food processing establishments with the ability to become a long-term resident of the facility. We have selected QAC, tertiary alkyl amine-, and chlorine-based disinfectants and tested their inactivating effect in two experimental set ups comprising a plate count method and a microscopic technique, *i.e.* EFM.

2. Materials and methods

2.1. Preparation of cell cultures

Pseudomonas aeruginosa ATCC®27853 and food isolates of *Staphylococcus aureus* FI 2 and *Listeria innocua* FI 2, obtained from Department of Industrial and Food Microbiology, Faculty of Food Science, University of Warmia and Mazury in Olsztyn, which revealed a high propensity for biofilm formation in a prescreen using a CV assay, were selected for this study. Individual *P. aeruginosa* ATCC®27853, *S. aureus* FI 2 and *L. innocua* FI 2 strains were grown in tryptic soy broth (TSB; Merck, Darmstadt, Germany) at 37 °C for 24 h. The cell cultures were adjusted to a cell density of 1×10^6 cells/mL in TSB and equal volumes were combined for preparing mixed species cell cultures of *P. aeruginosa* ATCC®27853-*L. innocua* FI 2 and *S. aureus* FI 2-*L. innocua* FI 2.

2.2. Formation of biofilms

The experiments comprising biofilm formation were performed according to (Olszewska et al., 2016). One milliliter of either single or dual (1:1) species cell cultures were transferred into 24-well flat-bottom polystyrene plates (Costar, Corning, NY, USA) and incubated at 30 °C for 72 h. At 24-h intervals, supernatants with planktonic cells were removed and biofilms were harvested by scraping the plate surface with a sterile polyester-tipped swab (Deltalab Eurotubo, Barcelona, Spain). Swabs were placed in glass tubes with 9 ml of saline (0.85% NaCl) and sonicated at 38 kHz (Elmasonic SB-120DN, Abchem, Warsaw, Poland) for 1 min and then vortexed for another 1 min. One hundred microliters of biofilm-released cells or their appropriate dilutions were spread plated on agar plates with CFC (Cephalothin, Fucidin, Cetrimide) (Biocorp, Warsaw, Poland), Baird-Parker-agar (Merck), and Oxford-agar (Biocorp) for enumeration of *P. aeruginosa* ATCC®27853, *S. aureus* FI 2 and *L. innocua* FI 2, respectively. The plates were incubated at 37 °C for 48 h. After incubation, colonies typical for each species were counted and the results were calculated as $\log_{10}$ CFU/mL. The mean values of viable bacteria cell counts ($\log_{10}$ CFU/mL) were obtained from at least four independent trials with two repetitions.

2.3. Inactivation of biofilms

To provide a larger surface for biofilm growth we used polycarbonate (PC) coupons with a diameter of 12.7 mm, selected from a variety of different coupons surfaces, for inactivation experiments. Single or dual species biofilms were developed on the coupons, which were placed in wells of 24-well polystyrene plates and filled with 1 mL of prepared cell cultures. Plates were incubated at 37 °C for 72 h. 24- and 72-h biofilms were treated with three disinfectants: quaternary ammonium compound- (Pursept AF, 2.0%), tertiary alkyl amine- (Barren, 2.0%) and chlorine-based (Medicarine, 0.18%) disinfectants. Sterile PBS (Sigma Aldrich, Poland) was used as a control. The growth medium was first removed and coupons were taken out with sterile forceps, washed with PBS and soaked in disinfectant solutions or PBS for 5 min. After treatments, coupons were removed and placed in Petri dish with D/E neutralizing broth (BD, Le Point de Claix, France). After

10 min the coupons were gently washed with PBS and transferred into Falcon tubes, suspended in 1 ml of saline (0.85% NaCl) and sonicated at 38 kHz (Elmasonic SB-120DN, Abchem, Warsaw, Poland) for 1 min and then vortexed for another 1 min. Finally, biofilm-released cells or their appropriate dilutions were spread plated on agar plates, as described above. After 48 h of incubation the colonies were counted and all results were expressed as log reductions (−). The mean values were obtained from at least four independent trials with two repetitions.

2.4. EFM with FISH and cell viability assay

Overnight cell cultures of individual strains at a cell density of 1×10^2 cells/mL were mixed in equal volumes in fresh TSB and added to sterile 4-well LabTek™ chamber slides (Thermo Fisher Scientific, Waltham, USA) at 800 µL per well. Slides were incubated at 30 °C for 72 h and growth medium was changed every 24 h to sustain bacterial viability. After incubation, the medium was removed and the biofilms were fixed with PBS-ethanol (1:1) at −20 °C for 30 min before fluorescence *in situ* hybridization (FISH) and then the procedure was conducted as described elsewhere (Olszewska, Kocot, & Łaniewska-Trokenheim, 2015). In this study, we used an oligonucleotide probe specific for *Listeria* species: 5′ CCC CAA CTT ACA GGC 3' (Brehm-Stecher, Hyldig-Nielsen, & Johnson, 2005), which was labelled with CY3 (cyanine 3) and DAPI (4′,6-diamidino-2-phenylindole) a nuclear counterstain to visualize the companion strain in the biofilm. The microscopic examination was performed with an epifluorescence microscope (Olympus BX51, Olympus, Hamburg, Germany), equipped with a 100 W mercury lamp, XC digital camera and a set of filters: U-M41007: 530–560 nm and U-MNU2: 360–370 nm for detection of orange and blue fluorescence, respectively. Prior to LIVE/DEAD staining, the wells were rinsed with PBS and subjected to disinfectant treatments as described above. Sterile PBS was used as a control. After treatments, the wells were rinsed with PBS and refilled with D/E broth for neutralization, incubated for 10 min and then rinsed again with PBS. Subsequently, the wells were filled with a staining mixture, which was prepared according to manufacturer's instructions and incubated at 37 °C for 30 min, in the dark. The staining mixture was then removed and biofilms were washed with PBS and mounted on glass slides underneath cover slips using immersion oil (Molecular Probes, USA). LIVE/DEAD viability kit contains two stains, Syto®9 and PI (propidium iodide), and allows differentiating live (green) cells from dead (red) cells based on cell membrane integrity. The microscopic examination was performed with an epifluorescence microscope (Olympus BX51, Olympus) equipped with a 100 W mercury lamp, XC digital camera and a set of filters: U-MNB2: 470–490 nm and U-MNG2: 530–550 nm for detection of green and red fluorescence, respectively. Thirty pictures were taken for each well to evaluate biofilm adhesion characteristics using a 9-stage adherence scale (Le Thi, Prigent-Combaret, Dorel, & Lejeune, 2001) with some modifications (Olszewska et al., 2016).

2.5. Statistical analysis

Statistical analysis was conducted with Statistica software ver. 13.1 (StatSoft Inc., Tulsa, OK, USA). Inactivated samples were tested by one-way ANOVA test. The differences were considered significant at $P < 0.05$ level of probability.

3. Results

3.1. Growth of individual species in biofilms

Results of cell growth of *P. aeruginosa* ATCC®27853 and *S. aureus* FI 2 with *L. innocua* FI 2 in dual species biofilms are presented in Fig. 1. The number of *P. aeruginosa* ATCC®27853 cells decreased from 9.8 to 9.1 log CFU/mL ($P < 0.05$), whereas that of *L. innocua* FI 2 decreased from 8.6 to 8.0 log CFU/mL ($P > 0.05$), indicating that *P. aeruginosa* ATCC®27853 predominated *L. innocua* FI 2 irrespective of incubation time ($P < 0.05$) (Fig. 1A). In turn, the number of *S. aureus* FI 2 cells decreased from 7.4 to 7.0 log CFU/mL ($P < 0.05$), and that of *L. innocua* FI 2 declined slightly from 7.5 to 7.4 log CFU/mL ($P > 0.05$), which shows that the bacterial population of *S. aureus* FI 2 was somewhat lower than *L. innocua* FI 2 ($P > 0.05$) (Fig. 1B). In single species biofilms, numbers of *P. aeruginosa* ATCC®27853 and *S. aureus* FI 2 cells increased from 7.3 to 8.5 log CFU/mL ($P > 0.05$) and from 7.7 to 8.6 log CFU/mL ($P > 0.05$), respectively (Fig. S1, Supplementary material), thus reaching lower ($P < 0.05$) and higher bacterial counts ($P > 0.05$) than in dual species biofilms. As indicated in our previous study, *L. innocua* FI 2 cell count in the single species biofilm decreased over the period of 72-h (data not shown), as it was shown here for dual species biofilm.

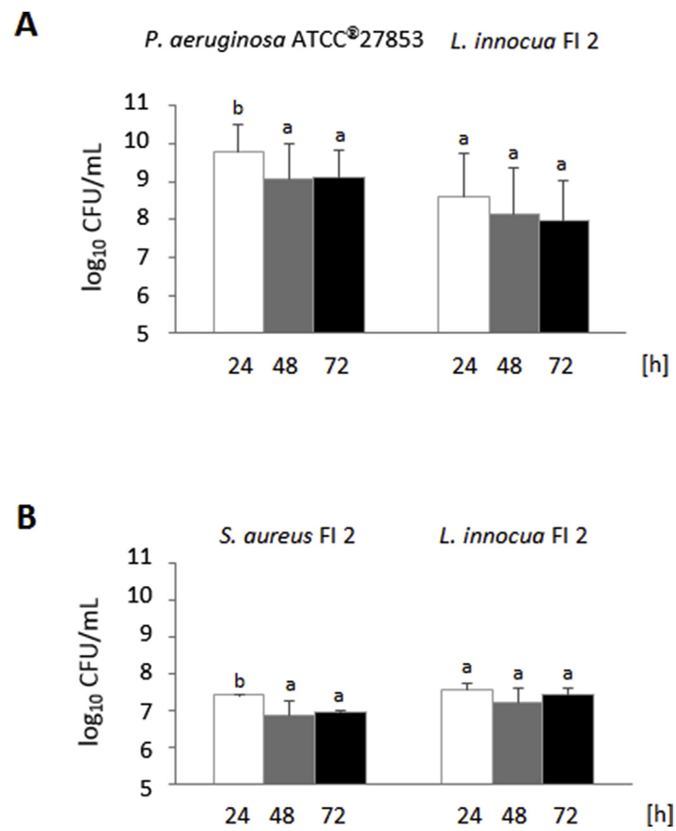

Fig. 1. The cell counts of *P. aeruginosa* ATCC®27853 − *L. innocua* FI 2 (A) and *S. aureus* FI 2 − *L. innocua* FI 2 (B) in mixed species biofilms formed in 24-well polystyrene plates at 24-h time intervals. The bars represent the mean values ± standard deviations. Different letters indicate significant differences among strains ($P < 0.05$).

3.2. Inactivation of cells in biofilms

Results of cell inactivation of *P. aeruginosa* ATCC®27853 and *S. aureus* FI 2 with *L. innocua* FI 2 in dual species biofilms are presented in Fig. 2. Distinct differences in cell inactivation were observed depending on the disinfectant used, with the QAC-based disinfectant being consequently the most effective.

Higher cell reductions of *P. aeruginosa* ATCC®27853 and *L. innocua* FI 2 were achieved in the 72-h dual species biofilms for each disinfectant used (Fig. 2A). The cell reduction magnitudes of 3.8 ($P < 0.05$), 1.1 ($P > 0.05$), and 0.6 ($P > 0.05$) log units between the 24- and the 72-h old biofilms were shown for *P. aeruginosa* ATCC®27853 after the treatment with QAC-, tertiary alkyl amine-, and chlorine-based disinfectants, respectively. In turn, magnitudes of 3.3 ($P < 0.05$), 2.0 ($P > 0.05$), and 0.8 ($P < 0.05$) log units between the

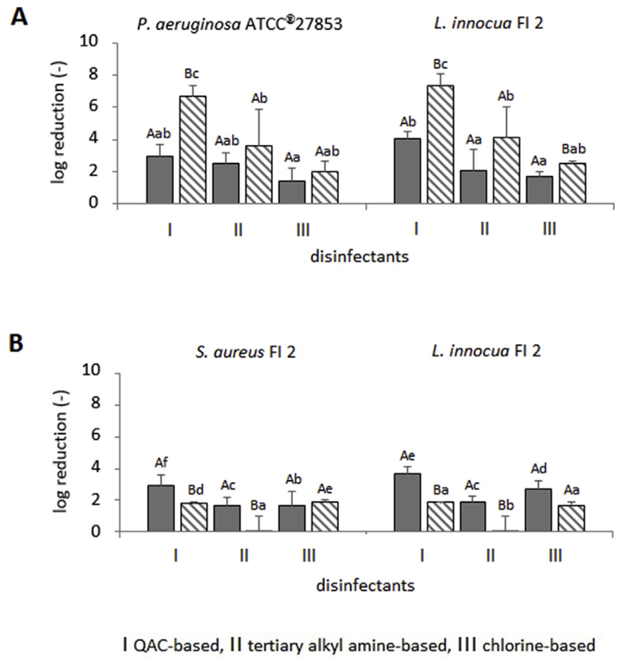

Fig. 2. The effect of three disinfectants on inactivation of *P. aeruginosa* ATCC®27853 – *L. innocua* FI 2 (A) and *S. aureus* FI 2 – *L. innocua* FI 2 (B) in 24- (filled bars) and 72-h (dashed bars) old mixed species biofilms after treatment for 5 min. The bars represent the mean values ± standard deviations. Different lowercase letters indicate significant differences among strains and different uppercase letters indicate significant differences between 24- and 72-h old biofilms ($P < 0.05$).

24- and the 72-h old biofilms were shown for *L. innocua* FI 2. *P. aeruginosa* ATCC®27853-*L. innocua* FI 2 were highly sensitive to the QAC-based disinfectant (cell reduction of 6.7 and 7.3 log units), and the least sensitive to the chlorine-based disinfectant, irrespective of biofilm maturity.

Higher cell reductions of *S. aureus* FI 2 and *L. innocua* FI 2 were observed in the 24-h dual species biofilms, except for the chlorine-based disinfectant which caused a greater cell reduction of *S. aureus* FI 2 in the 72-h old biofilm (Fig. 2B). The cell reduction magnitudes of 1.1 ($P < 0.05$), 1.6 ($P < 0.05$), and 0.2 ($P > 0.05$) log units between the 24- and the 72-h old biofilms were shown for *S. aureus* FI 2. In turn, cell reduction magnitudes of 1.8 ($P < 0.05$), 1.8 ($P < 0.05$), and 1.0 log units between the 24- and the 72-h old biofilms were demonstrated for *L. innocua* FI 2. Particularly noteworthy is the high resistance of both *S. aureus* FI 2 and *L. innocua* FI 2 to the tertiary alkyl amine-based disinfectant, as the cell reductions did not exceed 0.1 log units (Fig. 2B). In addition, *P. aeruginosa* ATCC®27853 and *S aureus* FI 2 were more resistant to the disinfectants tested in dual species biofilm than in the single species biofilms, except for *P. aeruginosa* ATCC®27853 which was more resistant to the QAC-based disinfectant in the 24-h single species biofilm (Fig. S2, Supplementary material). As for L. *innocua* FI 2, its higher resistance was demonstrated in 24-h dual species biofilms regardless of disinfectants used, whereas for *S. aureus* FI 2 higher resistance was observed in 72-h dual species biofilm.

3.3. EFM with FISH and LIVE/DEAD cell viability assay

Microscopic examinations of the *P. aeruginosa* ATCC®27853-*L. innocua* FI 2 biofilm after FISH, using a species-specific CY3 probe, resulted in red *L. innocua* cells, while DAPI staining resulted in blue cells of *P. auruginosa*. Microscopic examinations of the *P. aeruginosa* ATCC 27853-*L. innocua* FI 2 biofilm revealed both species contributed almost equally to the biofilm (Fig. 3A). As for *S. aureus* FI 2-*L. innocua* FI 2, we have observed red cells resulting from hybridization with the CY3 labelled probe (*L. innocua* FI 2) and blue cells from staining with DAPI (*S. aureus* FI 2); biofilms demonstrated a predominance of *S. aureus* cells (Fig. 3B).

Visual inspection of untreated biofilms stained with LIVE/DEAD® viability kit showed almost the entire visual field covered with cells or some free-of-cell areas that constituted for adherence stage no. 7 and 8 for *P. aeruginosa* ATCC®27853-*L. innocua* FI 2 and *S. aureus* FI 2-*L. innocua* FI 2, respectively, and that the cells had healthy membranes (Figs. 4A and 5A). In turn, cells with damaged membranes were observed in both dual species biofilms inactivated with the QAC-based disinfectant (Figs. 4B and 5B). In addition, in the *P. aeruginosa* ATCC®27853-*L. innocua* FI 2 biofilm this disinfectant caused dispersion of cells from the biofilm, leaving a few, small cell aggregates (stage 5), whereas in the *S. aureus* FI 2-*L. innocua* FI 2 biofilm cells still covered almost the entire visual field (stage 7). Following treatment with the tertiary alkyl amine-based disinfectant, a significant differentiation of cell populations into live and dead was observed (Figs. 4C and 5C). In *P. aeruginosa* ATCC®27853-*L. innocua* FI 2 biofilms, this disinfectant caused biofilm dispersal into a few, small aggregates and into single cells (stage 3), the majority of which maintained intact membranes (Fig. 4C). In the case of the *S. aureus* FI 2-*L. innocua* FI 2 biofilm, most of the visual field was covered with cells forming microcolonies, with small free-of-cell areas (stage 7) and a predominance of *S. aureus* FI 2 live cells (Fig. 5C). Following treatment with the chlorine-based disinfectant, only single cells with damaged membranes and a few aggregates (stage 3) were observed in the *P. aeruginosa* ATCC®27853-*L. innocua* FI 2 biofilm (Fig. 4D), whereas almost the entire visual field was covered with cells (stage 7) differentiated into live and dead cells in the *S. aureus* FI 2-*L. innocua* FI 2 biofilm (Fig. 5D). Also here, a predominance of *S. aureus* FI 2 cells was observed (Fig. 5D).

4. Discussion

Two experimental set ups were conducted to examine the effects of selected disinfectants on biofilm inactivation: one performed with PC coupons where quantitative data were derived from the plate count results, and the second one performed in a LabTek™ chamber slide system where qualitative data were derived from the FISH and LIVE/DEAD staining. We selected three disinfectants (QAC-, tertiary alkyl amine-, and chlorine-based) to evaluate their antimicrobial efficacies on dual species biofilms of *P. aeruginosa* ATCC®27853-*L. innocua* FI 2 and *S. aureus* FI 2-*L. innocua* FI 2 in comparison with single species biofilms.

The selected disinfectants showed different antimicrobial behaviors against the biofilms depending on their maturity. Current studies have reported that antimicrobial resistance is linked to biofilm maturity. For instance, Yang, Kendall, Medeiros, & Sofos (2009) showed an "older" biofilm formed by *L. monocytogenes* on high-density polyethylene cutting board surfaces to be more resistant to nine disinfectants than the "younger" biofilm, which could be due to the greater biofilm thickness developed over a longer period of time and to higher production of EPS surrounding the biofilm (Liu, Du, Zhao, Zhao, & Doyle, 2017; Uhlich, Cooke, & Solomon, 2006). In turn, Wang, Cai, Li, Xi, & Zhou (2018) did not observe any greater cell reduction along with extended exposure of *P. fluorescens* biofilm to chlorine dioxide (20 mg/L) even though the cell number decreased in time. This may be explained by the depletion of nutrients leading to eventual cell death, while dead cells form a barrier preventing antimicrobial agents' penetration into the biofilm. Similar observations were made by de Oliveiraa, Brugneraa, Cardosob, Alvesc, & Piccolia (2010), who extended the exposure time of essential oils of *Cymbopogon citratus*, which did not enhance their efficacy against *L. monocytogenes* biofilm. Our study showed a higher resistance of the 72-h old *S. aureus* FI 2-*L. innocua* FI 2 biofilm compared to the 24-h old biofilm, which may also be explained by the protective effect of dead cells, especially since the cell count in the biofilm incubated for 72 h was observed to decrease. Opposite observations were made for the *P.*

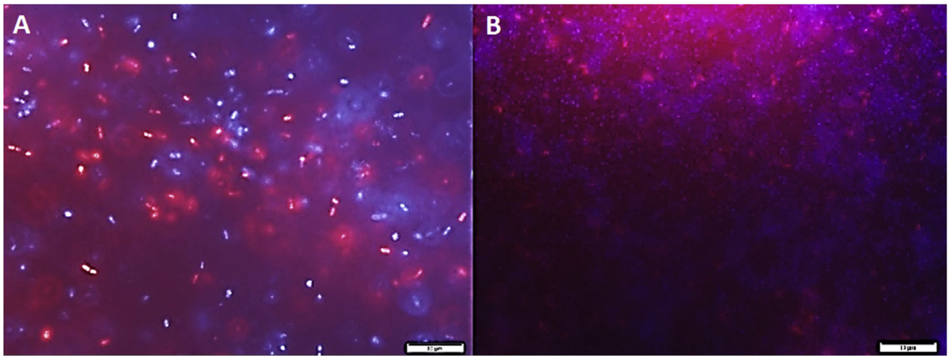

Fig. 3. Representative EFM images of *P. aeruginosa* ATCC®27853 – *L. innocua* FI 2 (A) and *S. aureus* FI 2 – *L. innocua* FI 2 (B) in 72-h mixed species biofilms formed in LabTek™ chamber slide system showing *P. aeruginosa/S. aureus* (blue) and *L. innocua* (red) cells visualized after FISH. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

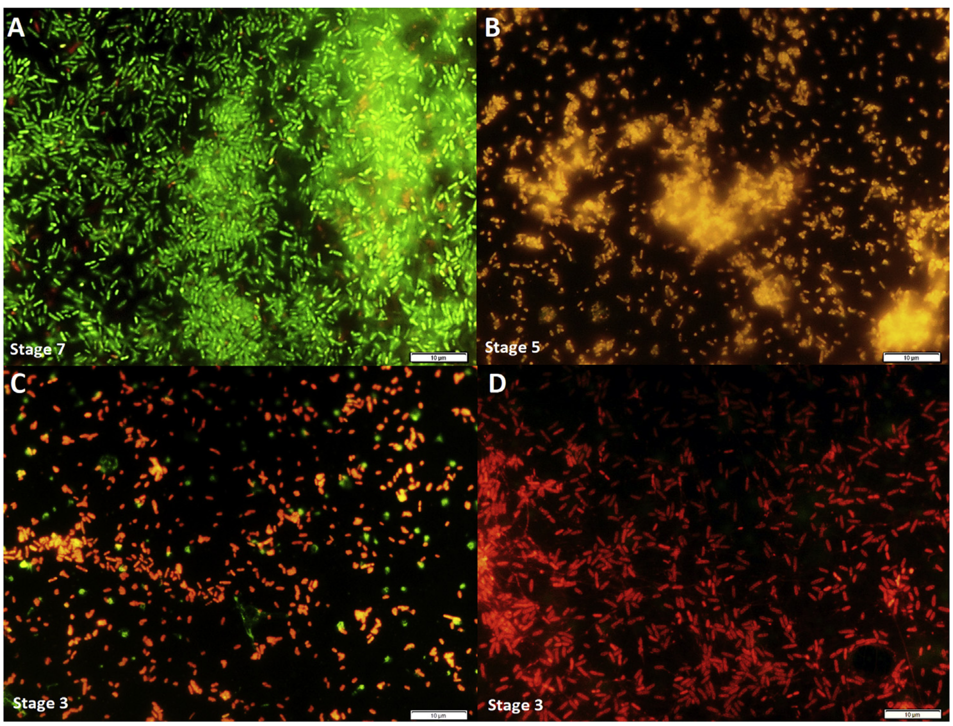

Fig. 4. Representative EFM images of *P. aeruginosa* ATCC®27853 – *L. innocua* FI 2 in 72-h old mixed species biofilms formed in LabTek™ chamber slide system and stained with LIVE/DEAD BacLight™. Images represent live (green) and dead (red) cells after 5 min treatment with disinfectants (A – untreated, B – QAC-based, C – tertiary alkyl amine-based and D – chlorine-based disinfectant), including stage of biofilm adherence as described in Materials and methods. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

aeruginosa ATCC®27853-*L. innocua* FI 2 biofilm which was more sensitive to the tested disinfectants in the 72-h biofilm than in the 24-h old biofilm. *P. aeruginosa* exhibits a propensity to produce EPS, which could impair the penetration of antimicrobial agents into the biofilm (Bridier, Briandet, Thomas, & Dubois-Brissonnet, 2011; Ma et al., 2009). However, *P. aeruginosa* is a quick biofilm producer, which allows it to quickly reach the mature 'mushroom shaped' form (Wang, Cai, LiXu, & Zhou, 2018). Most probably dispersal of bacterial cells occurred in the "older" biofilm, which was indicated by decreased cell counts in the 48- and the 72-h old biofilms. It may, hence, be speculated that the biofilms formed by quick biofilm producers, such as *P. aeruginosa*, rapidly attain the mature form and then undergo degradation. Thus, the weak cell organization in such biofilms provides poorer protection against antimicrobial agents. Despite ample research conducted into the resistance of *Pseudomonas* spp. biofilms to disinfectants (Fatemi & Frank, 1999; Giaouris et al., 2013; Saá Ibusquiza, Herrera, Vázquez-Sánchez, & Cabo, 2012), still sparse information is available on the antimicrobial behavior in dual and mixed species biofilms depending on the stage of biofilm maturity. Thus, further studies addressing the changes of biofilm architecture and cell organization over time are needed.

Many investigations addressing biofilms have been focused on single species biofilms compared with their planktonic counterparts (Aarnisalo et al., 2007; Amalaradjou et al., 2009; Cruz & Fletcher, 2011; Kim, Pitts, Stewart, Camper, & Yoon, 2008). Currently, more studies are focusing on dual and mixed species biofilms, which better mimics naturally occurring biofilms including those found in food processing environments. Previous studies have reported the mixed species biofilms to be more resistant to antimicrobial agents than the single species ones. For instance, Rodríguez-López et al. (2017) demonstrated that *L. monocytogenes* with *E. coli* was more resistant to benzalkonium chlorine (BAC), when the biofilms were formed on the stainless steel surface. The higher resistance of *L. monocytogenes* to BAC with *Pseudomonas putida* was also observed by Saá Ibusquiza, Herrera, Vázquez-Sánchez, & Cabo (2012). Van der Veen and Abee (2011) demonstrated that dual species biofilm of *L. monocytogenes-Lactobacillus plantarum* was more resistant to BAC and paracetic acid than their single species biofilms. Previous studies have shown that the higher resistance in dual species biofilms was observed also for other species. Behnke, Parker, Woodall, and Camper (2011) found that *P. aeruginosa* was more resistant to chlorine in a co-culture with *Burkholderia cepacia*, thus suggesting that interactions between two species affected their resistance. In turn, Millezi et al. (2012) studied the resistance of *S. aureus* and *E. coli* and showed that both of them were more resistant to lemon and citronella essential oils in the mixed species biofilm. Results of our study strongly correspond with previous findings, because we found that not only *L. innocua* FI 2 but also most common foodborne pathogens were more

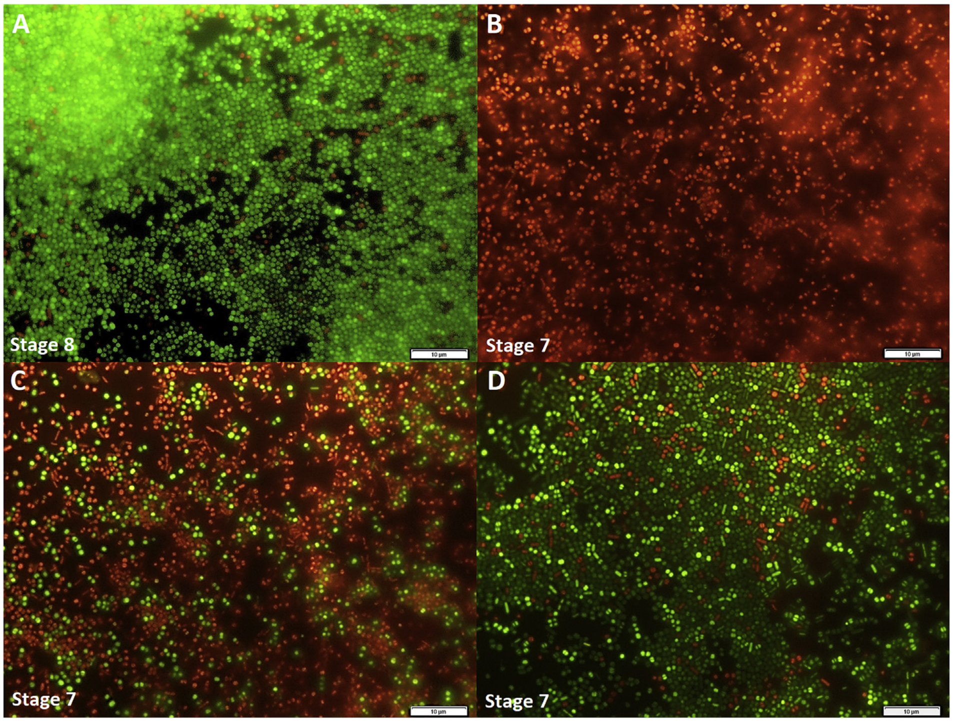

Fig. 5. Representative EFM images of *S. aureus* FI 2 – *L. innocua* FI 2 in 72-h old mixed species biofilms formed in LabTek™ chamber slide system and stained with LIVE/DEAD BacLight™. Images represent live (green) and dead (red) cells after 5 min treatment with disinfectants (A – untreated, B – QAC-based, C – tertiary alkyl amine-based and D – chlorine-based disinfectant), including stage of biofilm adherence as described in Materials and methods. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

resistant in the mixed species biofilm. Both *P. aeruginosa* ATCC®27853 and *S. aureus* FI 2 were more resistant to tertiary alkyl amine-, chlorine- and *S. aureus* FI 2 also to QAC-based disinfectants. The higher resistance of dual species biofilms is probably due to the interspecies interactions in biofilms and a key role may be played by *quorum-sensing*, chemical composition of EPS, and localization of individual strains in the biofilm, which will be investigated in our future studies.

In our study, we used EFM with LIVE/DEAD staining and FISH. The EFM-LIVE/DEAD enabled the qualitative analysis of cells after disinfectant treatments based on cell membrane integrity, whereas EFM-FISH revealed the microbial composition in *P. aeruginosa* ATCC®27853-*L. innocua* FI 2 and *S. aureus* FI 2-*L. innocua* FI 2 biofilms. For instance, the EFM-LIVE/DEAD allowed observing the efficacy of the tested disinfectants against biofilms and – importantly – demonstrated a destructive effect of the chlorine-based disinfectant on membrane integrity of *P. aeruginosa* ATCC®27853- *L. innocua* FI 2 biofilm cells, while majority of cells in the *S. aureus* FI 2- *L. innocua* FI 2 biofilm maintained their cell membrane integrity following the same treatment. Thus, when nutrients were repeatedly supplied for 72-h time period, dual species biofilm organization provided some protection to *S. aureus-L. innocua* but failed to protect bacteria when *P. aeruginosa-L. innocua* were organized in the biofilm. Hence, further studies will undoubtedly allow for better recognition of this phenomenon and for extending the still scarce knowledge on the interactions between cells and their effect on cell survival. Nevertheless these findings can be useful in the context of designing effective strategies to control bacterial contaminants in food processing plants, wherein bacteria often meet favorable conditions to organize themselves in biofilms, e.g. access to nutrients. This is particularly relevant in the case of pathogenic bacteria which can persist in the food processing environment for long periods especially in locations hardly accessible for cleaning and disinfection procedures. Biofilm set-up in which bacteria adopt multicellular behavior may facilitate their long-term survival and thus hinder the adjustment and performance of disinfectants commonly used in the food industry to eliminate bacterial contamination.

In conclusion, this study provides useful information about the antimicrobial effects of chemical disinfectants on *P. aeruginosa* and *S. aureus* with *L. innocua* in dual species biofilms depending on maturity and in comparison to single species biofilms. Demonstrating the enhanced resistance of selected species in dual species biofilms indicates the importance of the species composition and thus interactions in biofilms in their response to disinfectants. In addition, the efficacy of disinfectants can also depend on the stage of biofilm maturity and environmental conditions, *i.e.*, nutrient accessibility, hence they may indicate difficulties in the adjustment and performance of the applied disinfectant treatments. This may be of particular importance in the context of the long-term colonization of the food processing environment by microbes, especially foodborne pathogens posing a food safety risk.

Conflict of interest and authorship conformation form

Please check the following as appropriate:

- o All authors have participated in (a) conception and design, or analysis and interpretation of the data; (b) drafting the article or revising it critically for important intellectual content; and (c) approval of the final version.
- o This manuscript has not been submitted to, nor is under review at, another journal or other publishing venue.
- o The authors have no affiliation with any organization with a direct or indirect financial interest in the subject matter discussed in the manuscript

Declaration of competing interest

The authors declare no conflict of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.lwt.2019.108736.

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